

Opinion

Plant Phenomics—the unrecognized rise of a scientific discipline

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Plant Phenomics studies the phenotypic dynamics that form plant phenotypes by systematically measuring traits using phenotyping methods from the quantum to the ecosystem levels. It has emerged as an interdisciplinary field advancing sensing, computation, and plant biology. However, Plant Phenomics has lacked a unifying framework that integrates its community's core concepts from the formal and life sciences. Central to this framework is the definition of the phenome as a set of phenes that govern phenotypic dynamics across all spatial and temporal scales of biological and ecological organization and in interaction with the environment. This paradigm moves beyond gene-centric views and recognizes the equal importance of all spatial and temporal scales in forming plant phenotypes, advancing Plant Phenomics as a data-driven discipline and its emerging profession, the plant phenomicist.

The unusual origins of Plant Phenomics as a discipline

Previous research by Schneider has shown that scientific disciplines evolve sequentially over several phases [1]. Schneider introduced an evolutionary model that classifies a scientific discipline into one of four sequential Schneider stages. The first, or founding Schneider stage introduces a new object of inquiry as the result of initial discoveries and defines the core language and phenomena studied. The second Schneider stage focuses on method and tool development. Once methods are established, the third Schneider stage answers specific questions and generates fundamental insights. During the fourth and final Schneider stage, a discipline ceases to produce fundamental insights but preserves and broadly disseminates its knowledge.

In Shneiders model, Plant **Phenomics** (see [Glossary](#)) initially emerged as the technological and methodological support for scaling research in plant breeding [2] and plant physiology [3], two disciplines in the fourth and final Schneider stage. Inheriting their methodological framework, the Plant Phenomics community formed around sensor technologies and methods for high-throughput measurement of plant **traits** [4]. Yet this origin obscures what Plant Phenomics has since become.

Plant Phenomics is now a discipline that uses **phenotyping** methods to systematically decipher plant functions encoded in the dynamic changes in plant form and organization, that is, the plant **phenotype** [5]. Once plant phenotype is quantified, Plant Phenomics links functions to interactions between processes across **spatial** and **temporal scales** of biological and ecological organization [6] and to the natural and human-managed environment. These interactions are the **phenotypic** dynamics forming the measurable phenotype.

Highlights

Plant Phenomics lacks a unifying framework to fulfill the hallmarks of a scientific discipline. We propose a community-driven, multidisciplinary, quantitative consensus definition integrating core concepts from formal and life sciences, centered on the idea that processes across multiple scales jointly produce an observed phenotype.

Plant Phenomics explores the phenome as multiscale transformations (phenes) that interact with the environment across time and space to explain measurable phenotypes, requiring rigorous defining of the phenome to characterize phenotypic dynamics across scales and how they shape their environment.

We introduce the plant phenomicist as the profession of Plant Phenomics: experts who build formal systems to discover phenotypic dynamics, aiming to understand how plants thrive in any environment and solve grand challenges in agriculture and conservation.

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The need to measure phenotypes fostered the integration of sensors, data analytics, data management, and computational modeling as central tools for transforming large-scale phenotyping data into biological insights and economic value [7,8]. The increasing heterogeneity and complexity of data drove the adoption of artificial intelligence (AI) tools [9] and advanced mathematical and statistical methods [10] to push the frontiers of open data and automated analysis [11]. These methods empowered scientists and nonscientists alike to analyze large phenotyping datasets [12,13] and extract complex morphological features from 2D and 3D images. Consequently, it became possible to characterize plants and capture their surrounding environment [14], whether it is natural or human-managed, biotic or abiotic.

Building on these technological foundations, Plant Phenomics has produced landmark phenotyping methods to quantify plant phenotypes. These methods include characterizing gene–environment interactions with mathematical models to predict plant phenotypes [15] and latent space phenotyping to recover simulated quantitative trait loci from synthetic image datasets [16]. Another early focus was, high-throughput phenotyping in agricultural production environments. Early methods resolved and measured highly self-occluded branching structures, for example, in terrestrial laser scanning data to measure complete apple tree crowns in orchards [17] and increased throughput by introducing drones to phenotype whole orchards for chlorophyll content, leaf area index, and water stress, as exemplified by studies on olive and peach trees [18]. Moving into the soil, a key advance in Plant Phenomics is repeatable root phenotyping in the field [19]. As a result, the availability of all these sophisticated phenotyping methods then enabled the development of phenomic selection as an impactful applied method to increase yields through plant breeding [20].

The rise of landmark methods is indicative of the second Shneider stage, and one might ask: ‘Did Plant Phenomics skip Shneider stage one or only advance plant breeding and plant physiology?’ Our answer is no. These phenotyping methods (Figure 1, yellow line) also enabled a new mode of studying plants centered on phenotypic dynamics as the new object of inquiry. Initial studies (Figure 1, green line) found the genetic mechanisms underlying higher levels of biological organization. Examples include 2D image analysis to measure traits that link to quantitative trait loci governing morphological development in tomato fruits [21] and 3D laser ablation tomography to identify the anatomical traits that could be linked to the genes controlling the formation of multiseriate cortical sclerenchyma and its impact on soil penetration [22]. Other Plant Phenomics studies discovered plant survival mechanisms under abiotic stresses. These studies used cell shape analysis to unravel unknown epidermal root cell types that dynamically aid seedling establishment in response to the soil environment [23], employed robot-assisted time-series screening of whole plants to show that survival and growth are genetically decoupled under drought [24], and used whole-plant imaging of above- and below-ground organs to distinguish root–shoot growth dynamics between inbred and hybrid maize lines [25]. Phenotypic dynamics have also been observed through video demonstrating genotype-specific root interactions between two plants [26] and by measuring plant-emitted chemicals that encode inducible signals associated with partial resistance to root rot caused by microbes [27].

Here, we argue that Plant Phenomics has entered Shneider’s first stage, founding a discipline around phenotypic dynamics as a new object of inquiry. Yet Plant Phenomics’ origins in method development for plant breeding and plant physiology already equip the discipline with the cutting-edge technologies of a second-stage discipline. In the following sections, we describe a unifying multiscale framework for the **phenome** and characterize the emerging profession of the **phenomicist**, thereby positioning Plant Phenomics as the template for Phenomics throughout the biological sciences. On these grounds, we call for the recognition and support of Plant

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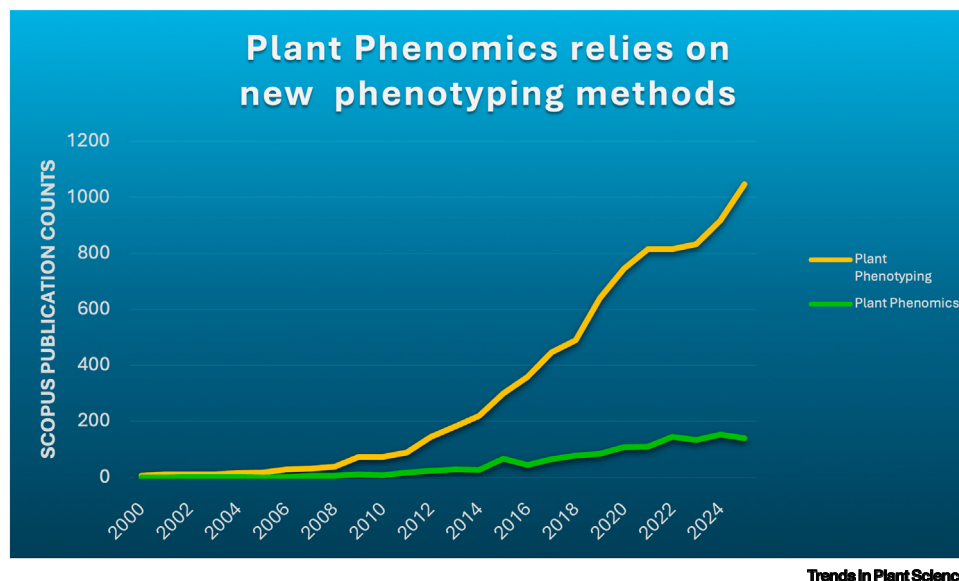
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Figure 1. Publications in Plant Phenomics trail the development of phenotyping methods. Plant Phenomics (green line) is the study of the plant phenotype. Over the past 2 decades, Plant Phenomics has intentionally centered on developing improved high-throughput plant phenotyping methods (yellow line) to quantify plant phenotypes in an automated, sensor-, and computer-assisted way, resulting in exponential growth in the number of phenotyping methods. Only recently have discoveries in Plant Phenomics begun to accelerate, driven by the rapid expansion of available phenotyping methods. The trend was quantified as the number of publications using each term ('Plant Phenomics' and 'Plant Phenotyping') in Scopus as of 27 November 2025. Phenomics and phenotyping are used interchangeably by some researchers; however, we assume that both terms are equally affected by inconsistent terminology.

Phenomics as a discipline by funders and scientific academies worldwide, to support Plant Phenomics on its way to reaching the third Shneider stage, where the majority of fundamental discoveries are being made.

The Phenomics framework of multiscale plant systems

'Plant Phenomics does not prioritize one spatial or temporal scale over the other and fundamentally acknowledges the importance and connectedness of all scales.'

Plant Phenomics acknowledges the complexity of plant phenotype characterization and the necessity of integrating data across spatial and temporal scales to advance our fundamental understanding of plants as living organisms interacting with and shaping their environments [28].

The multiscale framework offers a compelling approach to systematically capture the complex dynamics underlying phenotype formation. Within this framework, the phenotype is the integration of interactions between processes occurring across spatial and temporal scales of biological and ecological organization, from the quantum level [29] to entire ecosystems [30] and from fractions of a millisecond [31] to millennia [32]. The Phenomics wheel (Figure 2) illustrates that each spatial scale corresponds to a specific time frame at which biological and ecological processes occur [33,34].

The Phenomics wheel defines the plant phenotype as the result of interactions among processes at all spatial and temporal scales. This definition contrasts with the current notion that only processes at lower scales drive emergent phenotypes at higher scales. Instead, the Phenomics

Glossary

Formal science: an umbrella term for all sciences that use abstract constructs, such as mathematics, statistics, theoretical computer science, systems theory, and logic.

Formal system: a system that can be expressed using methods from the formal sciences.

Phene: the transformation of one or more unit traits into measurable traits.

Phenome: The set of phenes defining the formation of a measurable, multiscale phenotype.

Phenomicist: a professional who defines formal systems to discover the multiscale dynamics of organism–environment interactions.

Phenomics: the study of phenotypes.

Phenotype: an organism's measurable characteristics resulting from interactions among processes across all spatial and temporal scales.

Phenotypic: defining or forming a phenotype.

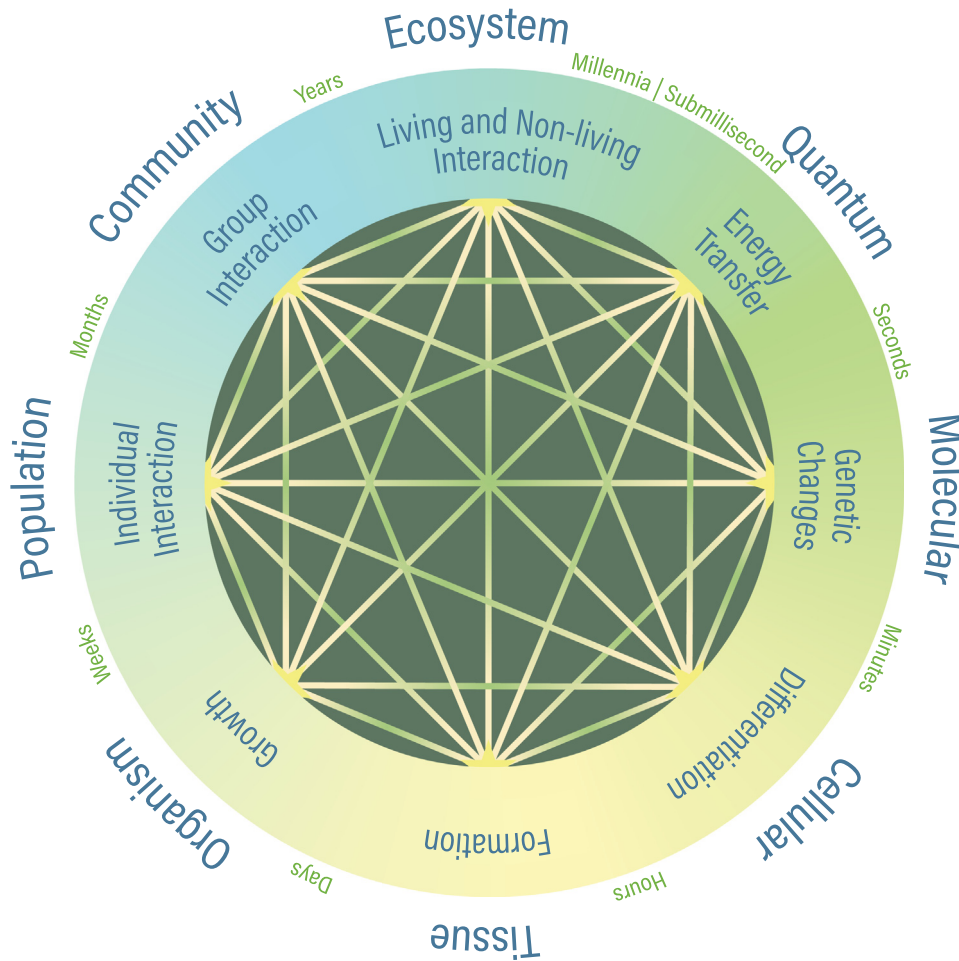
Phenotyping: methods and procedures for measuring and quantifying a phenotype as a set of traits.

Spatial scale: a physical area and level of detail through which a phenomenon is viewed. Each level of biological or ecological organization is associated with a spatial scale.

Temporal scale: a time window and level of time detail through which a phenomenon is viewed. Each level of biological or ecological organization is associated with a temporal scale.

Trait: a measurable quantity that characterizes an organism's phenotype.

Unit trait: a trait value has a unit magnitude of 1. The terminology is adopted from mathematics and denotes a scale-free quantity that can be transformed into any target number and associated with a unit.



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Figure 2. The Phenomics wheel of process interactions: The Phenomics wheel illustrates how processes (e.g., differentiation) at different spatial scales of biological and ecological organization (e.g., molecular) occur predominantly at specific temporal scales (e.g., days). The inner network represents the interactions between biological and ecological processes as edges. Spatial scales are denoted with blue labels, and temporal scales, positioned between the spatial scales, with smaller green labels outside the ring representing processes. Interactions across scales mean that changes at one scale can propagate to others, ultimately altering traits measured for a phenotype. Of note, the temporal ring has a break between submilliseconds at the quantum scale and millennia at the ecosystem scale. While the temporal scales form a singularity here, the processes impacting phenotype formation at the quantum and ecosystem scales interact.

multiscale framework posits that processes at multiple scales, both higher and lower, simultaneously contribute to the emergence of a phenotype. For example, a plant can induce shade avoidance responses, reflected in measurable differences in organ growth in a neighboring plant [35,36], thereby impacting tissue- and organ-level variation within a population. This simple example illustrates that considering all scales is essential, as phenotypes arise from interactions among all biological and ecological processes across spatial and temporal scales and with the environment.

Temporal scales indicate over which time frame environmental changes become measurable in the phenotype. These scales range from immediate reactions to long-term evolutionary

adaptations [37–39]. A mutation, a chemical reaction occurring over seconds, can lead to adaptations to degraded land over a millennium. Similarly, a plant leaf interacts with light within its ecosystem, where the spatial arrangement of the ecosystem, for example, mountains or other plants, determines the amount of light on the leaf. Light consists of quantum particles called photons that carry electromagnetic energy. When leaf pigments absorb photons, the resulting excitation energy is transferred through the light-harvesting complexes to drive photosynthesis [40]. Light availability in an ecosystem is one example of how ecosystem arrangement connects to quantum-level processes and can subsequently influence the growth of organs, such as roots and leaves. Embracing such multiscale interactions enables us to unlock the full potential of plants in any environment (Outstanding question 1).

These spatiotemporal interactions among processes constitute the phenotypic dynamics studied by Plant Phenomics, which can be observed by measuring the plant phenotype [41]. A deep understanding of these dynamics enables a quantitative assessment of a plant's potential to thrive in any environment [42,43]. Plant Phenomics fundamentally acknowledges the importance and interconnectedness of all spatial and temporal scales. Nevertheless, individual research efforts might prioritize a particular scale to further understanding of interconnected phenotypic dynamics (Outstanding question 2).

A foundational interdisciplinary theory of Phenomics

The formal Plant Phenomics setting

Plant Phenomics uses technologies to directly measure plant traits across spatial and temporal scales. Traits are the result of data processing, such as the brightness [44] and color [45] of images collected with cameras, plant height measured with rulers [46], levels of gene expression generated by a sequencer [47], and/or the absorption of X-rays by tissues [48].

Formally, a trait refers to a measurable quantity of the plant phenotype. Therefore, we define a 'trait value' T as the result of a direct measurement at a specific point in time t_i ($i = 0 \dots \infty$), arising from phenotypic dynamics. We can define t as the set consisting of all possible time points t_i on any time basis, such as real time, measured, for example, in hours; phenological time, measured as a sequence of developmental events; or evolutionary time, measured as the number of generations. The freedom to choose any time basis implicitly assumes that methods and technology exist to convert between the chosen time bases.

Mathematically, we represent phenotypic dynamics as transformations, called **phenes**, acting on **unit traits**. Here, 'unit' refers to the idea that a trait value has magnitude 1 and is associated with a measurement unit, a notion we adopted from mathematics to refer to a scale-free base value that is transformed into the measurable phenotype. As an oversimplified example, one can imagine a plant with rooting depth as the only trait. Rooting depth is denoted as 1 in S and is associated with the unit 'cm'. A phene is a transformation rule whose realized value depends on when and where it is evaluated. At day 5, the phene might generate a scaling transformation of 30 in a specific environment and result in a rooting depth of 30 cm. At day 10 and/or in a different environment, the same phene might generate a scaling transformation of 40, resulting in a rooting depth of 40 cm.

'The phenome quantifies the formation of an organism's multiscale phenotype'.

In the Plant Phenomics framework, phenes generate phenotypic dynamics that give rise to plant phenotypes and their associations with specific environments. Thus, a set N of m phenes O_m^N forms a phenotype P , characterized by measurable trait values $C_1, \dots, C_n$ in a measurable environment E at time point t_i . We remind readers that E encompasses both biotic and abiotic

environments, which may be naturally occurring and/or human-managed. In this sense, P can be understood as an ‘instance’ of the set N of m phenes, given the parameters of E as attributes. Consequently, the phenome consists of the set of phenes $N(O_1, \dots, O_m)$ that generate the phenotypic dynamics, giving rise to the possible phenotypes of a plant and their associations with environments.

Our definition integrates the distinction between measurements and observations from **formal sciences** that use abstract constructs to expand knowledge (e.g., mathematics, theoretical computer science, or statistics) into Plant Phenomics. This integration allows for the expansion of the $G \times E$ concept [49], where G denotes the ‘genetic’ effect on the phenotype, $\times$ denotes the interaction of G and E . The expansion replaces G by a set of unit traits across all spatial scales S . Accordingly, S includes G along with its measurable traits, such as gene expression, methylation, or promoter distance. More intuitively, S can be thought of as a set of traits one deems to be sufficient to capture the phenotype, yet without a scale.

The phenome $N(O_1, \dots, O_m)$ acts on the unit traits represented in S as a transformation that maps the unit traits S , given the environment E , into the phenotype P . Therefore, a phenome gives rise to a measurable multiscale phenotype P as a function of time t , consisting of all possible time points t_i , by transforming traits across the spatial scales of an organism, in conjunction with measurements of the environment (Figure 3).

$$P(t) = S \times N(t) \times E(t) \quad [1]$$

Equation 1 is directional because N is a parameterized transformation and reads from right to left as ‘ E acts on N as a parameterization, such that N acts on S as a transformation to generate the

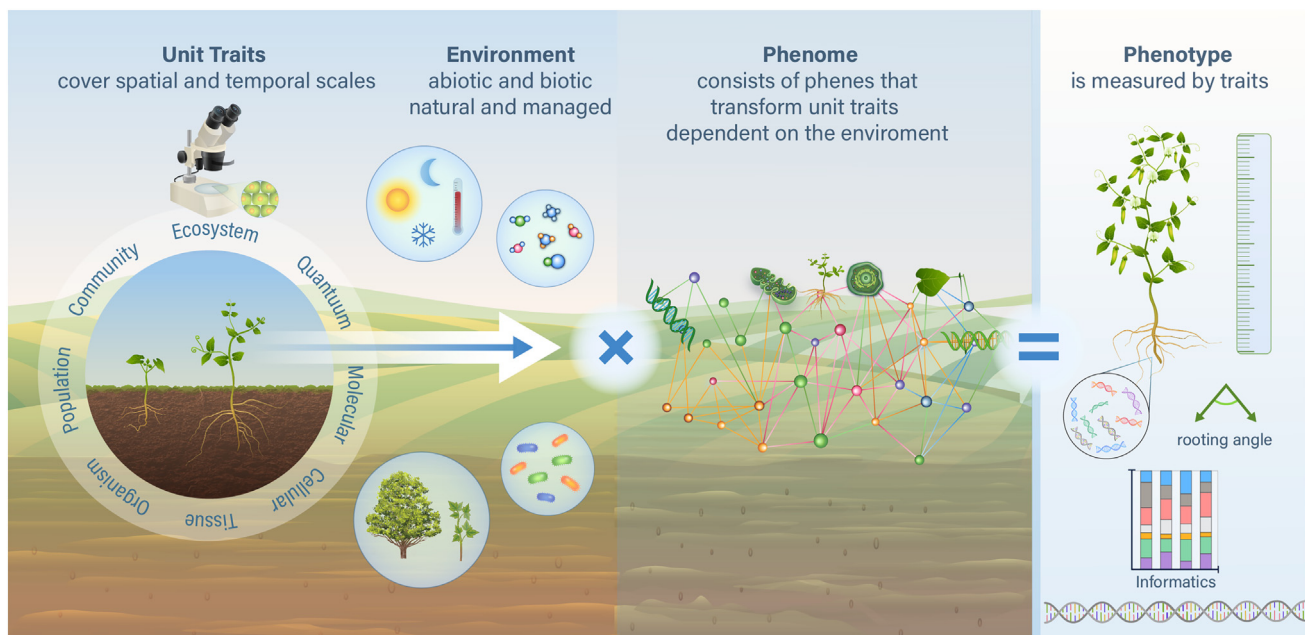


Figure 3. The Plant Phenomics concept visualized: The phenome quantifies how phenotypic dynamics form a plant phenotype as transformations of unit traits with respect to the environment. Traits characterizing the phenotype can be measured with tools ranging from simple rulers to state-of-the-art informatics, across spatial scales from the quantum and molecular scales through the organismal scale to the ecosystem scale.

measurable phenotype P .^{*} We note that, by expanding $G \times E$ in Equation 1, we implicitly expanded $P = G + G \times E + E$, where $+$ denotes the additive combination of effect terms because Equation 1 also expresses the additive effects on P as a special case, recovered when N depends affinely on E and acts affinely on S .

The definition of the phenotype in Equation 1 facilitates three important aspects that are unique to the Plant Phenomics community and contrasts with the widely used description of the phenotype $P = G + G \times E + E$:

i. Phenotypes transform measurable unit traits mathematically.

As transformation rules, phenotypes exist independently of the environment but facilitate dynamics among unit traits and with the environment when evaluated, thereby forming the phenotype. Phenomics is therefore defined as understanding the mathematical space of phenotypes, supported by experimental verification.

ii. Phenotype recognizes the multiscalarity of plants.

Phenotypes transform unit traits across the spatial scales of a plant into measurable traits within an environment, enabling the prediction of plant phenotypes and ‘plant–environment interactions’. The differences in phenotypic dynamics realized by the phenotype capture a plant’s plastic responses across environments and through time.

iii. Phenomics unifies mathematical and biological foundations.

The mathematical distinction between measurable traits, observable plant–environment interactions, and transforming phenotypes fosters the integration of the formal and life sciences into Plant Phenomics.

In this definition, a phenotype is a hypothesized yet quantitative transformation of unit traits that describes an observable plant–environment interaction. For example, gene expression, methylation, stomatal opening, root curvature, transpiration rate, and population size can be directly measured to characterize plant phenotypes. By contrast, water use efficiency, drought acclimatization, and adaptation are observable plant–environment interactions that cannot be directly measured. These are inferred from trait measurements that result from the phenotype acting on unit traits in the environment.

The key distinction from previous phenotype definitions is the differentiation between phenotypes, as transformations of unit traits that are associated with plant–environment interactions, and traits, as measurements of the phenotype. This distinction is essential because phenotypes capture the change in traits due to the environment rather than the trait itself. In that sense, the phenotype describes how the environment acts on organisms. This definition resolves issues in previous phenotype definitions that associated the phenotype with an infinite yet measurable trait space that is impossible to fully characterize [50].

The notion of an infinite trait set, however, contradicts other views of the phenotype in which phenotypes are the fundamental building blocks of the phenotype [51]. These concurrent ideas

* Notes

We note that $N(t)$ is formally an operator and would be denoted $P(t) = N(t, E(t)) \cdot S$. We also note that $\times$ denotes mathematically distinct interactions among terms in Equation 1 and in the classic $P = G + G \times E + E$ equation. However, we opted to keep notation that is intuitive to a general audience without a mathematical background.

postulate that phenes are measurable traits of the phenotype, analogous to genes being measurable building blocks of the genotype [52,53]. While insightful, this analogy does not fully hold, because physically existing genes can be absent from a genome, whereas the value of a trait is a nonphysical quantity that always exists, even if it is 0 or defined as infinite. Therefore, it remains a topic of scientific discourse whether a set of fundamental building blocks of the phenotype exists to characterize S . Although such a base of building blocks would be elegant, our phenome definition allows for independence from this question and is valid for any chosen trait set.

Separating phenes, as transformations, from traits, as measurable entities, defines an infinite yet continuous phenome space of transformations acting on unit traits S . As a result, the phenome can be interpolated between phenes, while any measurable trait value in a discrete set of traits remains achievable. Consequently, the phenome N can be studied to test a hypothesized trait base by fostering an understanding of how measured traits arise predictably, as well as the hypothesized phenes that transform the phenotype to explain measured traits (Outstanding question 3).

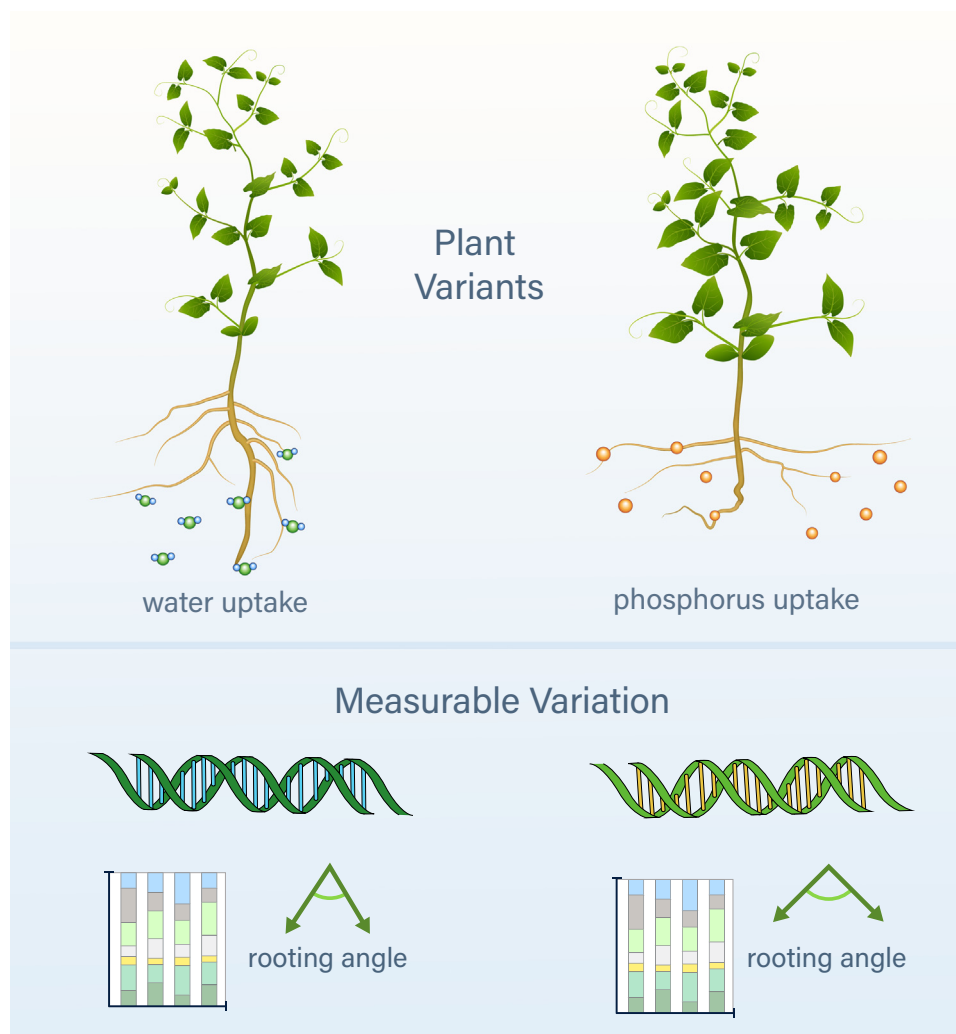
From a broader perspective, this definition of plant phenotypes formalizes why Plant Phenomics constitutes a distinct scientific discipline. The phenes summarized in N cannot simply be determined, as N also represents associations between phenes that often cannot be unambiguously decomposed. Consequently, the phenome N cannot be directly modeled or predicted with the tools available today. Instead, it requires a more comprehensive approach that uses the scientific method to explore and uncover the individual underlying phenes that form plant phenotypes and their observable plant–environment interactions. This distinction is crucial: it emphasizes that understanding the plant phenotype goes beyond measuring traits and demands a deeper, more methodical scientific inquiry into interactions across scales and with the environment, integrating perspectives from both the life and the formal sciences.

The Plant Phenomics concepts of variation and variants

Plant Phenomics treats variation as the variance observed in a measured plant trait at a specific time and in a specific environment. By contrast, we define a variant as the variance inferred at the phene level. In other words, a variant is associated with a plant's phenome, which reflects the distinct abilities of a plant to cope with a given environmental context and is measurable through trait variation.

For example, root morphological traits vary as a plant's phenotypic dynamics change. Assume Plant 1 and Plant 2 are in the same environment. Suppose Plant 1 has a deep root system, allowing it to avoid drought by accessing water in deeper soil layers. Likewise, Plant 2 can stimulate phosphorus mobilization through root exudates, which concentrate in the topsoil via a shallow root system, making it more effective at accessing phosphorus. To some extent, the two observable plant–environment interactions, 'responsiveness to drought' and 'responsiveness to phosphorus deficiency', can be measured using the trait rooting angle, quantified by a phene transformation. Therefore, depending on the context, the phenome of Plant 1 might be less efficient at extracting soil phosphorus, while Plant 2 might be better suited for low-phosphorus conditions and more susceptible to water deficit. Plant 1 and Plant 2 are then considered variants, although these two plants still show measurable variation in rooting angle (Figure 4).

A practical consequence of the Plant Phenomics framework is that variants are determined based on a cutoff point in trait variation. Therefore, variants are determined solely on the basis of a strong hypothesis, because N cannot be fully determined mathematically. Depending on the approach taken to model N , it is possible to define variants as instances of N , but this possibility is not explored in this article. Nevertheless, the scientific method is suitable for



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Figure 4. The distinction between phenotypic variants of plants and measurable trait variation. Plant 1 and Plant 2 represent phenotypic variants with beneficial resource-uptake characteristics suited to drought and phosphorus-deficient environments, respectively, realized through steep and shallow rooting angles. Both variants exhibit measurable variation in rooting angle.

exploring and confirming hypothesized phenes by integrating mathematical and experimental assessments of their heritability.

What is a plant phenomicist?

‘A plant phenomicist defines formal systems to discover the multiscale dynamics of plant–environment interactions.’

Ecologists explain interactions, physiologists explain mechanisms, geneticists explain inheritance, but phenomicists explain phenotypic dynamics that generate measurable phenotypes through interactions with the environment. Phenomicists strive to express objective reality, the underlying phenome, phenes, and variants, in terms of human-perceived reality—traits and

variation measured with phenotyping methods. This involves **formal systems** and their underlying mathematics to analyze and model phenotyping data that reflect the underlying phenotypic dynamics. There is, to date, no single training path to become a phenomicist, but phenomicists learn to work across disciplinary silos and to broaden their perspective on the use of data and models in different contexts of plant science.

Central to the profession is the formal investigation and understanding of phenes that orchestrate phenotypic dynamics across scales. This involves testing hypotheses and models about the existence of specific phenes using formal systems (Outstanding question 4). A formal system is a logical framework that derives conclusions by applying rules, technologies, procedures, and methodologies. This ranges from developing equipment, experimental designs, hardware, simulations and modeling, and software tools to evaluating across scales to understand the mechanisms underlying plant–environment interactions.

Through this formal study, plant phenomicists often aim to ‘saturate the (plant) phenome’ [54] as a set of traits for predicting a plant’s fitness, performance, and function under untested environmental conditions. However, understanding the mechanisms that give rise to phenomes consisting of phenes that orchestrate phenotypic dynamics is the grand challenge of Phenomics. A deep understanding of how a phenome forms and acts on unit traits across all scales will also allow the discovery of plant phenotypes that cannot be captured with data at the time of discovery (Outstanding question 5) and drive the development of management strategies that fully exploit a plant’s potential in any environment.

Plant phenomicists cannot be replaced by an AI agent in the near future

Plant Phenomics fundamentally strives for data-driven discovery: phenomicists formulate hypotheses about phenes that, by definition, are not yet characterized and may lack precedent across existing domains. Current AI systems excel at recognizing patterns within established knowledge but cannot originate the cross-domain hypotheses that define new phenes or propose phenotyping methods for dynamics that have not yet been observed. In this sense, AI is a powerful tool within the phenomicist’s formal system to accelerate pattern detection, automate trait extraction, and confirm hypothesized phenes against large datasets. Yet it does not replace the scientific reasoning required to conceive those hypotheses in the first place (Outstanding question 6). If anything, AI models are a reference of human knowledge that a phenomicist uses to ensure that discovery adds significantly to human knowledge.

Plant Phenomics drives technological innovations and standards toward a complete understanding of the phenome

Technologies for exploring the plant phenome are rapidly advancing, alongside a global network of large-scale infrastructures for plant phenotyping [55]. AI can predict phenotypes and help confirm phenes. Advances in quantum computing promise to enable probabilistic exploration of variations and variants at large scales, opening new avenues for understanding plant diversity. These tools also allow scientists to formulate and test hypotheses about underlying phenes and their resulting dynamics (Outstanding question 7).

Alongside rapid technological advancements, the Plant Phenomics community has developed robust data standards that support the FAIR (Findable, Accessible, Interoperable, and Reusable) principles [56] and the CARE (Collective Benefit, Authority to Control, Responsibility, and Ethics) principles [57]. Most notably, the community established the MIAPPE (Minimum Information About Plant Phenotyping Experiments) standard to ensure consistent metadata across experiments [58–60] and ontologies, capturing intricacies across all scales [61] to enhance data

interoperability. BrAPI (Breeding Application Programming Interface) is a software interface designed to interoperate with genotyping and phenotyping data standards in plant breeding [62,63], and OpenAlea [64] developed data structures for plant morphology across spatial and temporal scales [65,66], enabling interoperability between phenotyping and modeling software, such as PlantCV [67]. The cyberinfrastructure CyVerse [68], originally initiated by the National Science Foundation as a platform for plant biology [69], has enabled thousands of plant phenomicists to share their data and software and to stimulate the development of best practices for phenotyping data [70].

Plant Phenomics was established by its active community

‘Plant Phenomics explores the blueprint of plant life as the phenotypic dynamics across all spatial and temporal scales.’

The Plant Phenomics community has decisively established itself as an essential scientific discipline, and the presented Phenomics framework uses plants as a scalable model for all of biology. Plant Phenomics explores the blueprint of plant life as the phenotypic dynamics across all spatial and temporal scales interact with environmental factors, forging a unified theoretical framework and creating a language to define a phenome. Notably, the community has established the intellectual rigor necessary for recognition as a discipline by uniting perspectives from the formal and life sciences in its core concepts and by setting data, methods, and technology standards.

As a result of the community’s research activity, hundreds of publications on Plant Phenomics studies and plant phenotyping methods (Figure 1) have been disseminated through Plant Phenomics conferences and journals, such as *Plant Phenomics* (2019) and *The Plant Phenome Journal* (2018). As a professional organization, the International Plant Phenotyping Network serves as an umbrella organization for 18 national and regional networks documenting ongoing institutionalization. These networks organize a community spanning all career stages and include stakeholders from research institutions, breeding and high-tech companies, farmers, gene banks, and society [71]. As a consequence, the structure and activity of the Plant Phenomics community reflect the recommendations of global science advisory groups: The Science and Technology Action Committee [72], the Science Europe position statement [73], and Australia’s National Science and Research Priorities [74] emphasize collaboration across borders while fostering a diverse, skilled domestic workforce and nurturing international talent across science, technology, engineering, and mathematics (Outstanding question 8).

Plant Phenomics plays a critical role in addressing the grand challenges of the 21st century

‘The moonshot of Plant Phenomics is to understand and enable a plant’s full potential in any environment to provide plant yields and services anywhere, anytime, and for everyone.’

Realizing the moonshot to understand and enable a plant’s full potential in any environment to provide plant yields and services anywhere, anytime, and for everyone faces a grand challenge: a complete understanding of the phenome—how phenes generate phenotypic dynamics across spatial and temporal scales and how to quantitatively characterize an organism’s interactions with natural and managed environments in both biotic and abiotic scenarios.

The impact of Plant Phenomics is broad and profound, with every step taken toward comprehensively understanding the plant phenome and accelerating our ability to deliver new plant

variants across a broad scope of impact ranging from agriculture through environmental science to climate resilience. Ensuring food security and nutrition, providing energy, serving as construction materials, generating chemicals for pharmaceuticals, and supplying raw materials for textiles all benefit from a deeper understanding of how plant phenotypes form and function. Furthermore, plants provide essential services for human life, including oxygen production, shade, wind diffusion, improved air quality, and carbon dioxide sequestration, all of which must be understood in the context of how plant phenotypes interact with their environment, communities and populations, and respond across their own spatial and temporal scales.

It is time for the scientific community, funders, and policymakers to recognize Plant Phenomics as a discipline

This is a critical moment for science policy to act and create dedicated programs within funding agencies and foundations to support this rapidly growing field. We also issue a call to global scientific academies to formally recognize Plant Phenomics as the template for a broader Phenomics discipline, akin to precedents such as Systems Biology, Sustainability, and Bioinformatics. Supporting Plant Phenomics means establishing the example of reaching the third Shneider stage for Phenomics across all of biology. The third Shneider stage is critical because the vast majority of discoveries are made during this stage. Yet it will take dedicated long-term efforts to implement these discoveries to achieve societal impact. For maize production, such long-term efforts are beginning to pay off [75], but other plant species will need to follow this example. Therefore, this support is the necessary next step to fully leverage the power of plant science as a scalable model for biology, accelerate our response to climate change, and unlock the solutions required for a food-secure future and a sustainable Earth ecosystem in which humans can thrive (Outstanding question 9).

Author contributions

A.B. led the development of the manuscript. Y.S.C., J.C., S.G., P.v.G., W.G., J.K., S.P., E.P., S.Sa., and S.Sh. are members of the International Plant Phenotyping Network Working Group 'Phenomics as a Science'. All other authors contributed critically to the opinion and commented on the manuscript drafts.

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Outstanding questions

Does a complete understanding of the phenome demand the discovery of 'quantum traits'? The curiosity answer is certainly yes, and we can hardly imagine which insights phenomicists will reveal. However, the practical answer to this question will be given by quantum biologists and their ability to develop reliable tools to design phenotypes through quantum state manipulation.

To what extent are trait values at one scale predictive of other scales? The answer to this question could enable the estimation of genomics information from trait values at the organismal scale or predict trait occurrence across space and time.

How can we quantify the full potential of a plant in any environment as a set of phenes? This question is the key step toward developing approaches that let plants thrive in any environment and a complete understanding of the phenome is the grand challenge to solve.

Do phenes possess physical manifestations? Genes were once a concept before they were physically discovered. Phenomes are multiscale transformations of unit traits that might have physical manifestations such as gene regulatory networks, chemical signaling cascades, or even a currently unknown manifestation.

Which phenotypes are unpredictable from measurement data? The phenome defines a space of possible phenotypes far larger than what we can currently observe and capture in available environments. As such, it may hold solutions we cannot yet imagine for the environmental scenarios of the future.

How to increase the number of plant phenomicists? The single most important aspect might be that the phenomics community is diverse and transdisciplinary, making phenomics jobs less likely to be eliminated or replaced by AI and therefore attractive to the next generation of scientists.

Can phenomes be confirmed across all environments worldwide to deepen our understanding of plants?

Declaration of interests

The authors declare no competing interests.

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Confirming phenes is not a purely scientific endeavor: it also requires international funding mechanisms and FAIR data policies that ensure data quality across the large datasets produced by specialized phenotyping infrastructures globally. Such coordination is necessary to run large-scale experiments with the massive data exchange needed to confirm general phenes.

Can we accelerate and broaden the public discourse to unite phenomicists as advocates for recognizing phenomics as a scientific discipline? Cultivating a first generation of scientists who identify as phenomicists instead of the discipline of their initial training will be key to elevating phenomics to the status of a scientific discipline.

What are the necessary steps to develop phenomics into a discipline at its own right? The next step is to fund and recognize phenomics as a scientific discipline such that a feedback loop between research and education emerges that drives phenomics discovery, discourse, and development.

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